# Assignment stage 1 description

- Due No due date
- Points 0

## Project Structure

- **Group component:** 3–4 students.
- **Individual component:** Each student selects and analyses their own dataset.

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# Stage 1 (20 marks, due Week 5)

The objective of Stage 1 is to explore data, define a research question, and demonstrate both **collaborative and individual analytical skills**.

## Components

This stage consists of:

## 1. Group Component (12 marks, ~1000 words)

The group will:

- Select a **common problem domain** (e.g., healthcare, finance, education, social media, etc.).
- Define a **shared research/business problem.**
- Conduct a **joint literature review.**
- Propose a **general analytical approach/framework.**

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## 2. Individual Component (8 marks, ~500 words per student)

Each student must:

- Select their **own dataset** relevant to the group's problem domain.
- Define **individual research question(s)** based on their dataset.
- Perform **independent exploratory data analysis (EDA).**
- Prepare and describe their own data.

# Report Structure (Combined Submission)

## Length

- Total: ~2500–3000 words.
- Group sections plus clearly identified **individual sections for each student.**
- Appendix, visualisations, and references excluded from word count.

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# Section 1: Problem (Group)

- Describe the problem from a general perspective.
- Highlight the business/research need.
- Define the **overall project objective.**
- Briefly indicate how individual datasets contribute to the problem.

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# Section 2: Literature Review (Group)

- Summarise **6–8 relevant papers.**
- Link literature to your proposed problem.

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# Section 3: Approach (Group)

- Describe the **overall analytical approach/framework.**
- Outline the techniques and models planned for Stage 2.

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# Section 4: Individual Data & Analysis (Individual – clearly separated per student)

Each student must include a **clearly labelled subsection**:

## Student Name: [Name]

- Dataset description (source, size, features)
- How the data was acquired
- Individual research question(s)

- Exploratory data analysis (EDA)
- Key insights (supported by 1–2 figures or tables)
- Data preparation (cleaning, transformation, sampling)
- Tools used

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# Marking Breakdown (20 marks)

## Group Component (12 marks)

- Problem definition & relevance – 3 marks
- Literature review quality – 5 marks
- Proposed approach – 2 marks
- Report Structure and Style – 2 marks

## Individual Component (8 marks)

- Dataset relevance and selection – 1 mark
- Research question clarity – 1 mark
- EDA and insights – 4 marks
- Data preparation and tools – 2 marks

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# Submission Guidelines

- One group report submission.
- Individual sections must be clearly labelled with student names.
- Use the headings specified above.
- Minimum line spacing: 1.15.
- Minimum font size: 10 pt.

| Criterion | Marks | Applies To | Notes / Alignment |
|---|---|---|---|
| **Problem Definition & Relevance** | 3 | Group | Clearly describe the problem and business/research need. State the overall project objective and motivate Stage 2. |
| **Literature Review** | 5 | Group | Summarise 6–8 relevant papers. Evaluate the quality and relevance of the sources. |
| **Proposed Approach** | 2 | Group | Explain the intended approach, methods, and plan for Stage 2. |
| **Dataset Relevance & Selection** | 1 | Individual | Each student selects an appropriate dataset relevant to the group problem. |
| **Research Question Clarity** | 1 | Individual | Research question(s) are clear, specific, and aligned with the dataset. |

| Criterion | Marks | Applies To | Notes / Alignment |
|---|---|---|---|
| **Exploratory Data Analysis (EDA) & Insights** | 4 | Individual | Present data exploration, visualisations, summaries, and key insights. |
| **Data Preparation & Tools** | 2 | Individual | Describe cleaning, transformation, sampling, and the tools/software used. |
| **Report Structure & Style** | 2 | Group | Ensure clear organisation, appropriate formatting, and academic writing quality. |
| **Total** | **20** | | |

**Stage 1 Rubric (group & ind - 20 Marks)**

| Criteria | Ratings | | | | Pts |
|---|---|---|---|---|---|
| Problem Definition & Relevance (Group)<br>threshold: 3.0 pts | **3 Pts**<br>**Excellent (3 pts): Problem is clearly defined, highly relevant, and well justified. Overall objective is clearly stated and motivates Stage 2.** | **2 Pts**<br>**Good (2 pts): Problem and objective are mostly clear and relevant, with some justification.** | **1 Pts**<br>**Satisfactory (1 pt): Problem is vaguely defined or lacks clear motivation.** | **0 Pts**<br>**Unsatisfactory (0 pts): Problem is unclear or missing.** | 3 pts |
| Literature Review (Group)<br>threshold: 5.0 pts | **5 Pts**<br>**Excellent (5 pts): Summarises 6–8 high-quality papers and critically links them to the problem.** | **3 Pts**<br>**Good (3–4 pts): Reviews relevant papers with adequate discussion and connections.** | **2 Pts**<br>**Satisfactory (1–2 pts): Limited coverage or weak links to the problem.** | **0 Pts**<br>**Unsatisfactory (0 pts): Inadequate or missing literature review.** | 5 pts |
| Proposed Approach (Group)<br>threshold: 2.0 pts | **2 Pts**<br>**Excellent (2 pts): Clear and appropriate analytical approach, methods, and Stage 2 plan.** | **1.5 Pts**<br>**Good (1.5 pts): Approach is mostly clear with minor omissions.** | **1 Pts**<br>**Satisfactory (1 pt): Basic approach with limited detail.** | **0 Pts**<br>**Unsatisfactory (0 pts): Approach is unclear or missing.** | 2 pts |
| Dataset Relevance & Selection (Individual)<br>threshold: 1.0 pts | **1 Pts**<br>**Excellent (1 pt): Dataset is highly relevant and appropriate for the problem domain.** | **0.5 Pts**<br>**Satisfactory (0.5 pt): Dataset is somewhat relevant but limited.** | **0 Pts**<br>**Unsatisfactory (0 pts): Dataset is inappropriate or not described.** | | 1 pts |
| Research Question Clarity (Individual)<br>threshold: 1.0 pts | **1 Pts**<br>**Excellent (1 pt): Research question(s) are clear, specific, and aligned with the dataset** | **0.5 Pts**<br>**Satisfactory (0.5 pt): Questions are somewhat clear but lack specificity.** | **0 Pts**<br>**Unsatisfactory (0 pts): Questions are unclear or missing.** | | 1 pts |
| Exploratory Data Analysis (EDA) & Insights (Individual)<br>threshold: 4.0 pts | **4 Pts**<br>**Excellent (4 pts): Thorough EDA with meaningful visualisations, summaries, and** | **3 Pts**<br>**Good (3 pts): Good EDA with appropriate visualisations and reasonable insights.** | **2 Pts**<br>**Satisfactory (1–2 pts): Basic EDA with limited analysis or interpretation.** | **0 Pts**<br>**Unsatisfactory (0 pts): Little or no EDA performed.** | 4 pts |

| Criteria | Ratings | | | | Pts |
|---|---|---|---|---|---|
| | **insightful observations.** | | | | |
| Data Preparation & Tools (Individual)<br>threshold: 2.0 pts | **2 Pts**<br>**(Individual) Excellent (2 pts): Appropriate cleaning, transformation, and tools are clearly described and justified.** | **1.5 Pts**<br>**Good (1.5 pts): Most preprocessing steps and tools are described.** | **1 Pts**<br>**Satisfactory (1 pt): Limited preprocessing or insufficient explanation.** | **0 Pts**<br>**Unsatisfactory (0 pts): No evidence of preparation or tools used.** | 2 pts |
| Report Structure & Style (Group)<br>threshold: 2.0 pts | **2 Pts**<br>**Excellent (2 pts): Well-organised, professionally written, and follows formatting requirements.** | **1.5 Pts**<br>**Good (1.5 pts): Mostly clear and organised with minor issues.** | **1 Pts**<br>**Satisfactory (1 pt): Adequate structure but inconsistent presentation.** | **0 Pts**<br>**Unsatisfactory (0 pts): Poorly structured or difficult to follow.** | 2 pts |
| | | | | | Total points: 20 |